P3.py

```
# import lib
import  pandas as pd
from sklearn.linear_model import LinearRegression

# load data
data = pd.read_csv("housing.csv")
print(data)

# clean data
def  split_info(data):
        data[["area", "bedrooms"]] = data["info"].str.split(",", expand=True)
        data["area"] = data["area"].astype(float)
        data["bedrooms"] = data["bedrooms"].astype(float)
        return data[["area", "bedrooms"]]

# features and target
features = split_info(data)
target = data["price"]

print(features)
print(target)

# model
model = LinearRegression()
model.fit(features.values, target)

# prediction
area = float(input("enter area "))
bedrooms = float(input("enter number of bedrooms "))
test = pd.DataFrame({"info" : [str(area) + "," + str(bedrooms)] })
price = model.predict(split_info(test))
print(price[0])
```

P4.py

```
# import lib
import  pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.pipeline import  make_pipeline
from sklearn.preprocessing import FunctionTransformer

# load data
data = pd.read_csv("housing.csv")
print(data)

# clean data
def  split_info(data):
	data[["area", "bedrooms"]] = data["info"].str.split(",", expand=True)
	data["area"] = data["area"].astype(float)
	data["bedrooms"] = data["bedrooms"].astype(float)
	return data[["area", "bedrooms"]]

# features and target
features = data[["info"]]
target = data["price"]

print(features)
print(target)

# model
model = make_pipeline(FunctionTransformer(split_info), LinearRegression())
model.fit(features, target)

# prediction
area = float(input("enter area "))
bedrooms = float(input("enter number of bedrooms "))
test = pd.DataFrame({"info" : [str(area) + "," + str(bedrooms)] })
price = model.predict(test)
print(price[0])
```